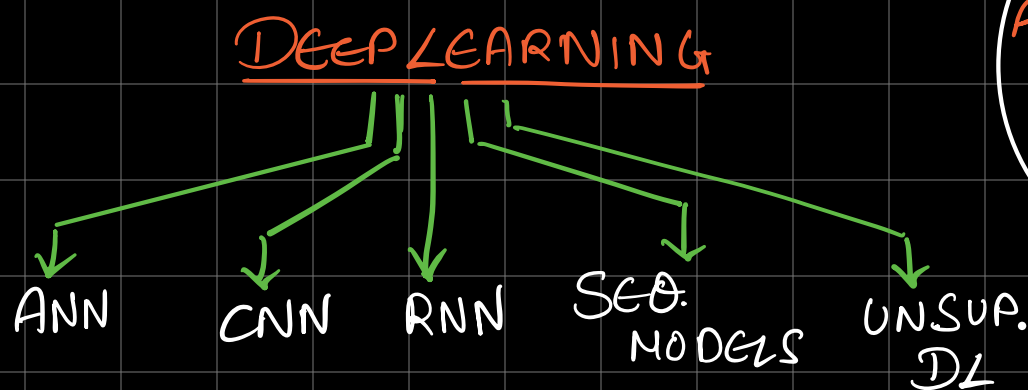


GEN-AI

It is a type of AI that creates new content by learning from existing data, mimicking human creativity.



SEQUENCE TASKS:

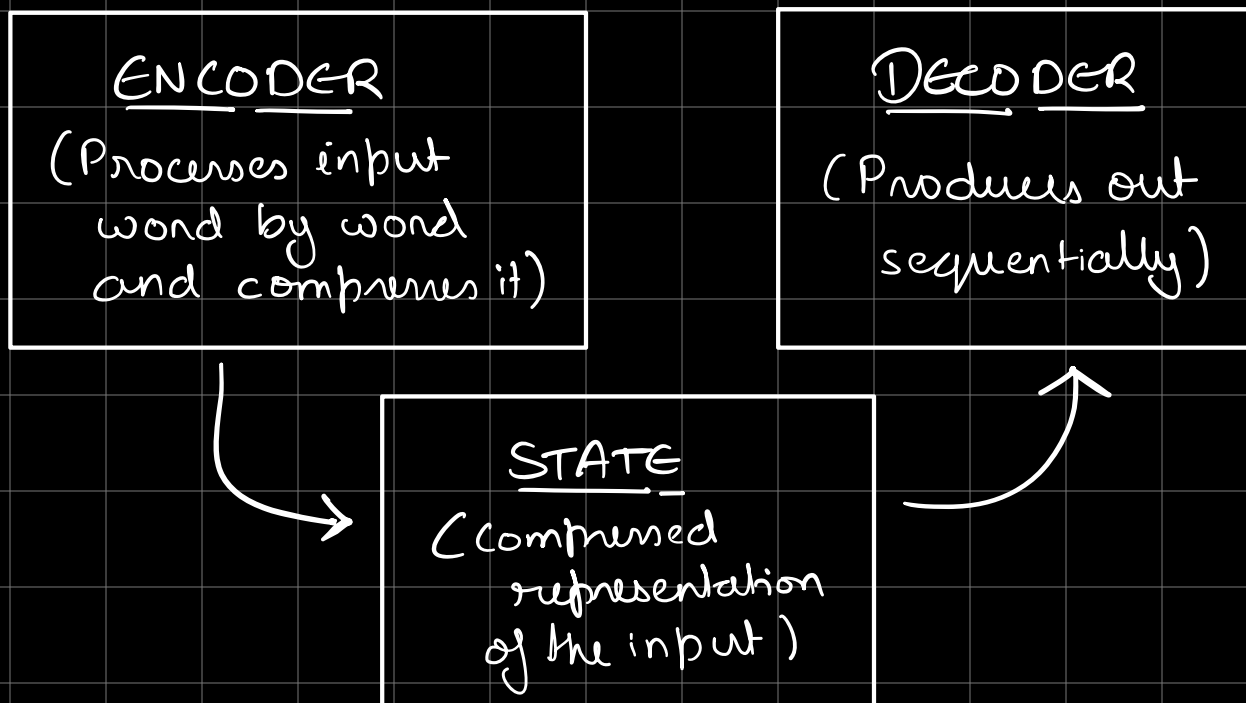
- 1) MANY TO ONE (Sentiment analysis)
- 2) ONE TO MANY (Image description)
- 3) SYNCHRONOUS SEQ TO SEQ (Parts of speech recog)
- 4) ASYNCHRONOUS SEQ. TO SEQ. (Translation)

WAYS TO SOLVE SEQ TO SEQ:

- 1) ENCODER-DECODER
- 2) ATTENTION MECHANISM
- 3) TRANSFORMER
- 4) TRANSFER LEARNING

5) LLM's

ENCODER-DECODER



⇒ Worked good for small input.

⇒ Problems with memory loss.

ATTENTION MECHANISM:

⇒ Has a similar encoder block but the decoder block is different.

⇒ All intermediate states are available unlike encoder-decoder which only has final state.





Attention layer is a neural network that finds the most relevant state for the given output of the decoder.

⇒ The training time / computational complexity of the attention mechanism is really high.

TRANSFORMER:

- ⇒ It still has encoder & decoder.
- ⇒ It can read all of inputs content all at once (Parallel Processing).
- ⇒ It can be trained in fraction of time as compared to previous arch.
- ⇒ The only drawback was that it needs a lot of data and it cannot be trained on small amounts of data.

TRANSFER LEARNING:

- ⇒ In this knowledge learnt from a previous task

can be used to boost performance on a related task.

LLM's:

⇒ Trained on huge datasets which made them extremely efficient for any task.

⇒ Hardware heavy

⇒ Works on transformer model